

HAT-P-18b

R-2474

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-18_180723 · created 2026-08-23T23:37:00+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2458322.78208 +/- 0.00018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.082 +/- 0.039
Transit depth (Rp/Rs)^2	0.67 +/- 0.64 [%]
Orbital Inclination (inc)	89.89 +/- 1.93
Airmass coefficient 1 (a1)	1.024 +/- 0.017
Airmass coefficient 2 (a2)	-0.0184 +/- 0.0073
Scatter in the residuals of the lightcurve fit is	1.63 %
Best Comparison Star	#2 - [244, 307]
Optimal Method	PSF photometry
Transit Duration (day)	0.095 +/- 0.029

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R* fit vs published	0.082 ± 0.039 vs 0.1356 (deviation 1.36σ)
Duration fit vs published	0.095 d vs 0.1129 d (deviation 0.61σ)
Mid-time O-C	-0.1 min
Depth SNR	2.1
Residual scatter	1.63 %

Light curve

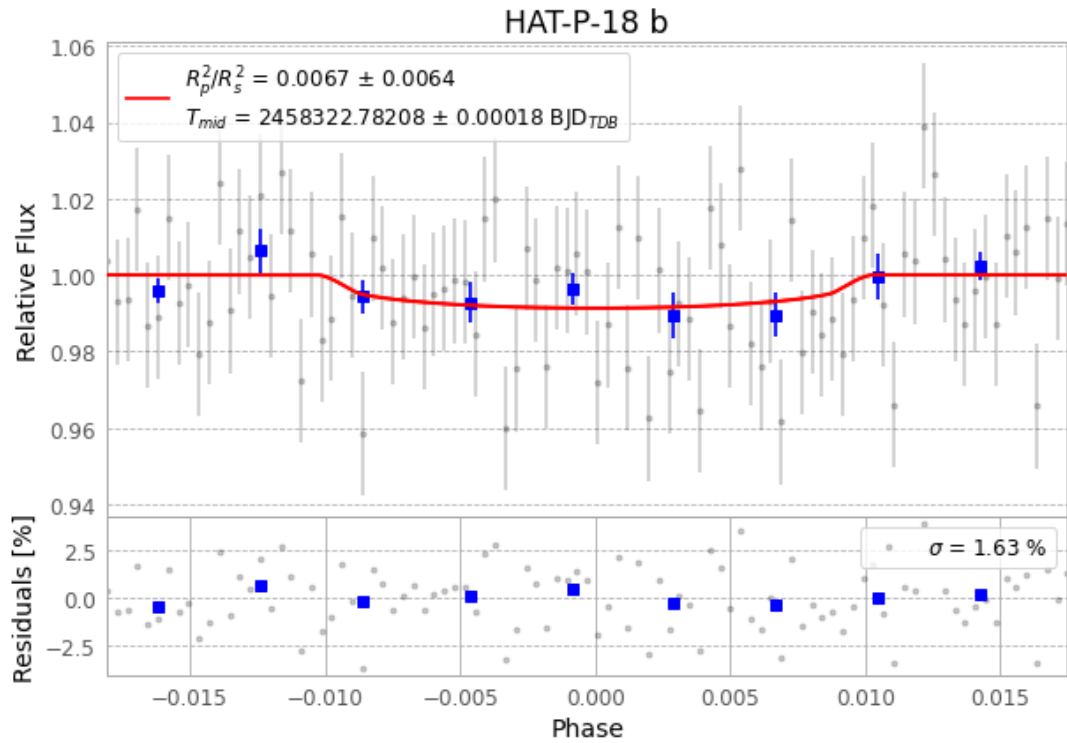


Figure 1 — FinalLightCurve_HAT-P-18 b_2018-07-22.png (SHA-256 a420f1135989e15f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [289, 118], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 7a22b792f2334ab7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

95 FITS frames (62 MB), HATP-18180723041812.FITS → HATP-18180723090012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-18-180723.log (f8c2518ddd37...), FinalLightCurve_HAT-P-18 b_2018-07-22.png (a420f1135989...), FinalLightCurve_HAT-P-18 b_2018-07-22.csv (a381e5276806...)

Compute resources

Wall time 128.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 23:37Z.